DATA 3464: Fundamentals of Data Processing

# Fancy feature transformations

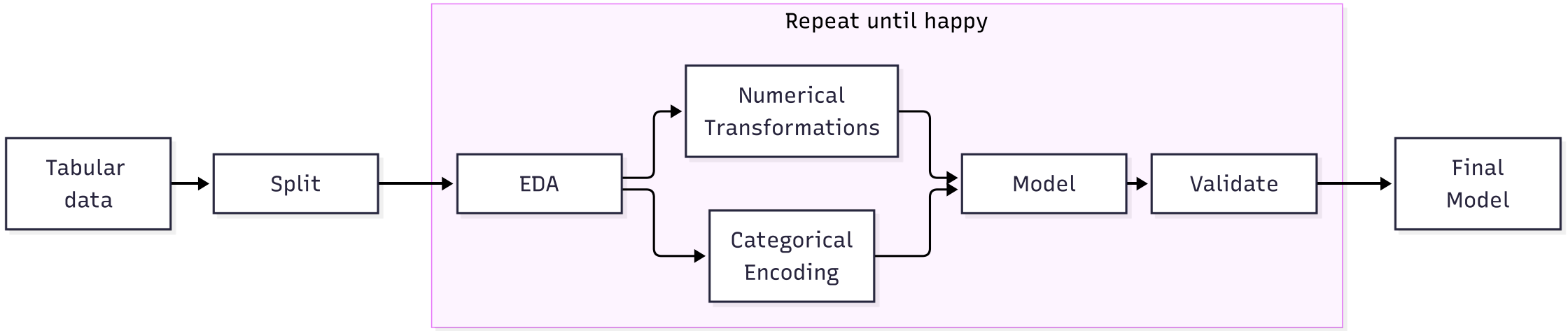
Charlotte Curtis

October 22, 2026

# Topic overview

- More complex categorical encoding
  - High cardinality
  - Infrequent categories
- More complex numerical transformations
  - 1:many
  - many:many

# The data process to date



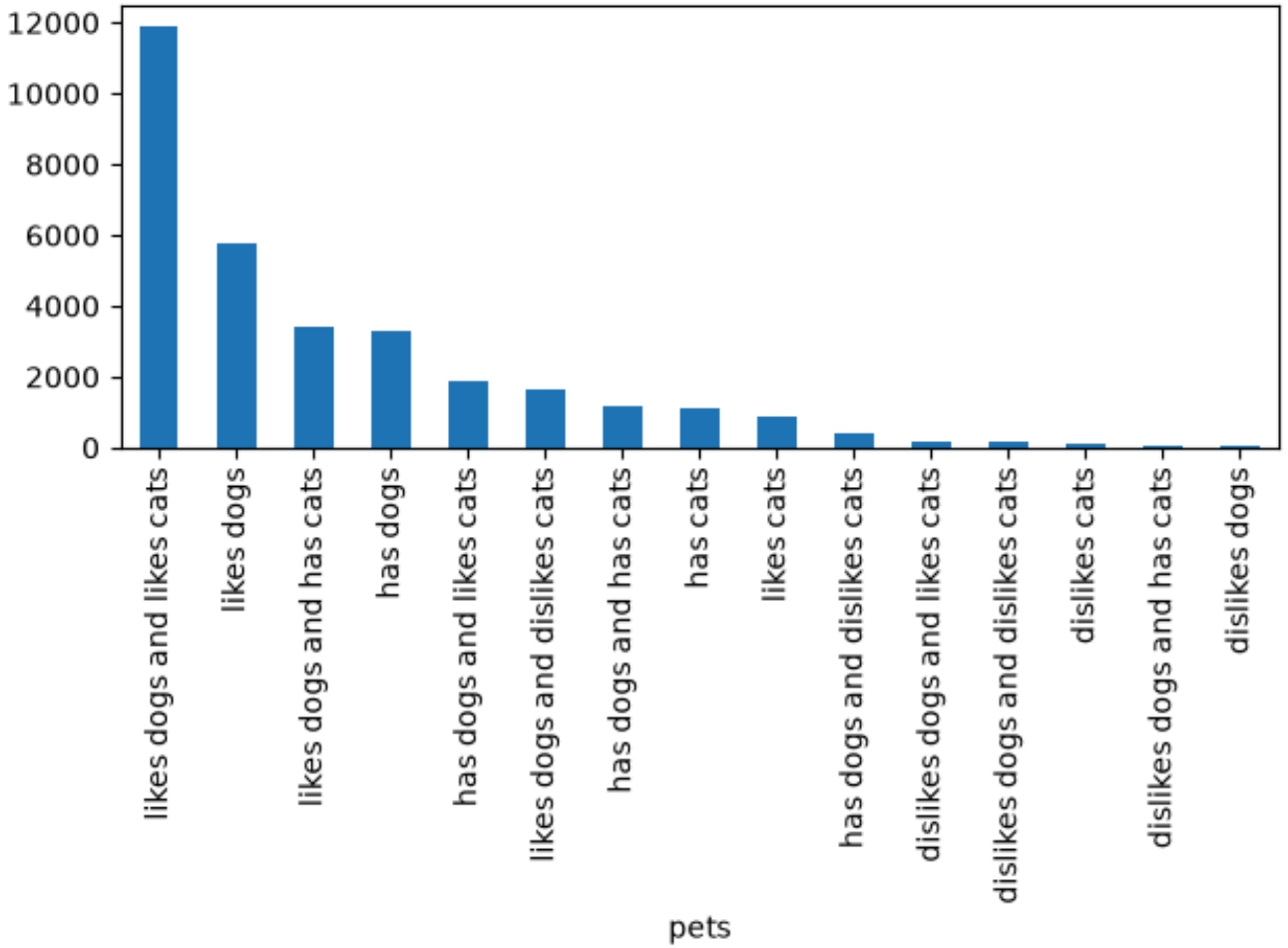

- We've covered one-hot and ordinal encoding
- We've covered 1:1 numerical transformations

> *Sometimes getting fancy is the answer*

# Potential problems with one-hot encoding

- One-hot encoding creates one column per category
- High cardinality implies some categories are infrequent
- Extreme case: $N$ samples have $N$ unique categories
- Very sparse matrix can cause numerical stability, high computation

> *Either we make it ordinal somehow, or we need to combine categories*




# Combining categories

- Even without high cardinality, might want to merge low frequency categories
- Pass `max_categories` as a constraint to `OneHotEncoder`

# Another approach: target encoding

- Basic concept: replace the category with the mean of the target
- Must be very careful during training to avoid data leakage!
- Example: predicting weight of animal

| Animal | | mean_kg |
|---|---|---|
| cat | → | 4.1 |
| dog | → | 15.4 |
| rabbit | → | 2.2 |

# Getting fancy with feature hashing

- Good if you have too many categories, or combinations of categories
- Converts each category into a fixed-length feature vector

| Animal | | A_0 | A_1 | A_2 | ... | A_16 |
|---|---|---|---|---|---|---|
| cat | → | 1 | 0 | 0 | ... | 1 |
| dog | → | 0 | 1 | 0 | ... | 1 |
| rabbit | → | 1 | 0 | 1 | ... | 0 |

> *Even fancier methods exist, like supervised encoding methods which basically train a model-within-a-model*

# Advantages/disadvantages

## Target encoding

+ Powerful if strong predictor
+ Compact

- Doesn't work for unsupervised
- Central tendency poor measure for categories with few samples
- Difficult to deal with novel categories

## Feature hashing

+ Natively handles novel categories
+ Compact (fixed columns/features)
+ Good for high cardinality

- Risk of hash collision
- Loss of interpretability/meaning

# Fancy numeric transformations

# 1:many transformations

- So far our numeric transformations have been 1:1
- 1:many creates multiple features from 1, like binning
- **Basis expansion** introduces nonlinearity to a continuous feature, e.g.:

$$x \rightarrow [\beta_1 x, \beta_2 x^2, \beta_3 x^3]$$

- More exciting is the **spline** version where you can define different polynomials for different ranges of $x$

Full disclosure: I have never used this approach

# many:many transformations

- Linear projection methods create new features:

$$X^* = XA$$

  where $A$ is the *projection matrix* and $X^*$ is the transformed data

- We can use this for **dimensionality reduction** by only keeping some of $X^*$
- Example: Principle component analysis (PCA) finds the set of orthogonal vectors that explain the most variance

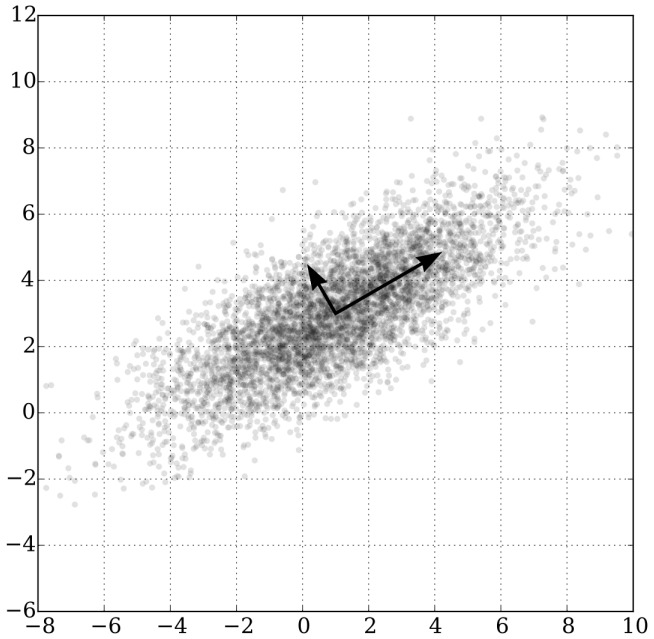



Figure from Wikipedia

# Variance and covariance - Review (?)

- The **variance** of a single feature $\mathbf{x}$ is:

$$\mathrm{var}(\mathbf{x}) = \frac{\sum_1^n (x_i - \mu_x)^2}{n}$$

- The **covariance** between two features $\mathbf{x_1}$ and $\mathbf{x_2}$ is:

$$\mathrm{cov}(\mathbf{x_1}, \mathbf{x_2}) = \frac{\sum_1^n (x_{1i} - \mu_{x_1})(x_{2i} - \mu_{x_2})}{n}$$

- Independent variables will have low covariance, but low covariance does not necessarily mean independence!

Or divide by *n-1* for an unbiased estimate of the population variance

# Covariance matrix

- The **covariance matrix** between all features in a matrix $X$ is:

$$\text{cov}(X) = \begin{bmatrix} \text{var}(\mathbf{x_1}) & \text{cov}(\mathbf{x_1}, \mathbf{x_2}) & \cdots & \text{cov}(\mathbf{x_1}, \mathbf{x_m}) \\ \text{cov}(\mathbf{x_2}, \mathbf{x_1}) & \text{var}(\mathbf{x_2}) & \cdots & \text{cov}(\mathbf{x_2}, \mathbf{x_m}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(\mathbf{x_m}, \mathbf{x_1}) & \text{cov}(\mathbf{x_m}, \mathbf{x_2}) & \cdots & \text{var}(\mathbf{x_m}) \end{bmatrix}$$

- Just like regular variance, covariance scales with the data
- If you normalize this, you get the **correlation matrix**

# Back to PCA

- Principle component analysis, proposed in 1901 by Karl Pearson, is a linear projection of multidimensional data onto the axes of maximum variance
- The axes are found by **eigendecomposition** of the correlation matrix:

$$C = Q\Lambda Q^{-1}$$

- The eigenvectors $Q$ form the new orthogonal basis for the correlation matrix $C$, while the eigenvalues $\Lambda$ represent the amount of variance "explained"
- $X^*$ can be calculated as:

$$X^* = XQ$$

# Dimensionality reduction with PCA

- Sort in order of decreasing eigenvalue ("amount of variance explained")
- Keep only the first $k$ eigenvectors to form $Q_k$
- Project data onto this smaller basis:

$$X^* = XQ_k$$

- Implemented in Scikit-learn by passing `n_components` to `PCA()`, e.g.

```
pca = PCA(n_components=2) # just two axes
X_reduced = pca.fit_transform(X)
```

# Linear discriminant analysis (LDA)

- Similar to PCA, but **supervised**
- Finds basis that maximize **class separability**
- Number of projection vectors must be `<= min(n_classes - 1, n_features)`
- Still uses projections and eigenvalues, but adds some statistical magic
- Key assumptions:
  - Features are normally distributed within a class
  - Classes have identical covariance matrices
- Implemented in Scikit-learn as LinearDiscriminantAnalysis

# Other methods

- It seems crazy, but random projections can work to reduce dimensionality
- Another many:many transformation that has gained popularity in the "Deep Learning era" is the **autoencoder**
- These are (typically) neural networks that learn a lower-dimensional representation by training the output to match the input
- The "bottleneck" layer in the middle is the reduced representation
- During inference, the model is chopped in half

# Pros and cons of dimensionality reduction

## Pros

- Can improve model by:
  - removing noise
  - reducing collinearity
- Lower complexity/compute cost
- Can help with visualization
- Can be informative about features

## Cons

- Throwing out information
- Harder to interpret results
- May not actually improve anything

> *Higher list of pros, but still... don't do it unless it's demonstrably beneficial*

# Coming up next

- Lab: build a model with the housing data
- Assignment 2: EDA, I look forward to seeing what you discover
- Lecture: Feature interactions